UK Atomic Energy Authority

UCL

simvue

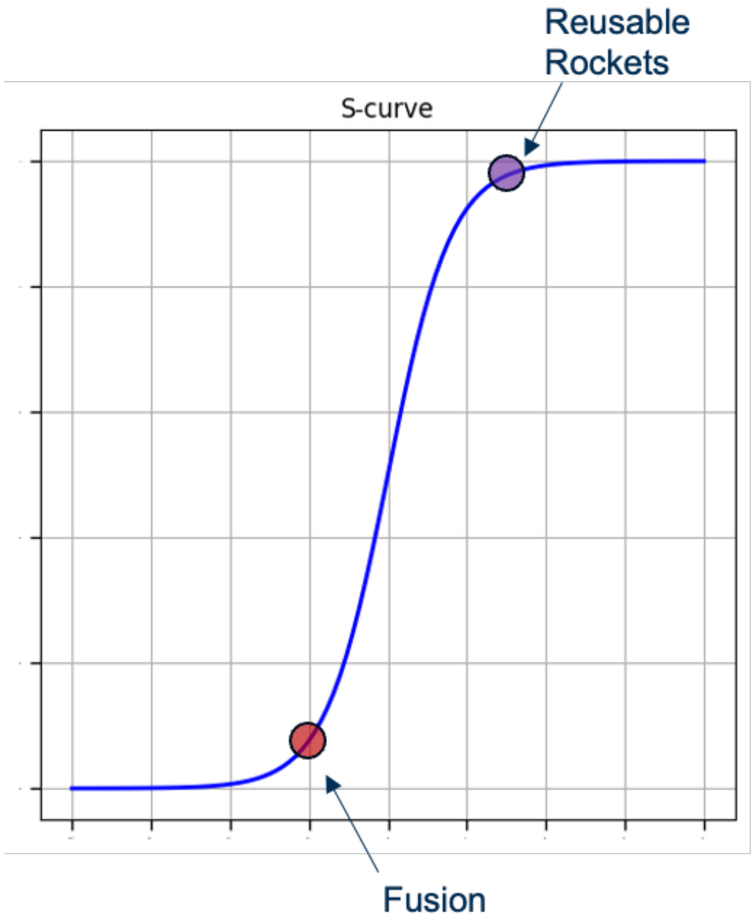
Learning Physical Operators using Neural Operators

A physics-informed approach to general-purpose neural operators.

Vignesh Gopakumar

International Conference on Scientific Computing and Machine Learning, Bath
15th September, 2026

State of Fusion Research



- Costly to use a design-build-test-iterate model
- Cannot experimentally observe reactor-grade burning plasma in current devices
- Will have to go up the engineering S-curve by going in-silico

Simulate everything



Even at the exascale?





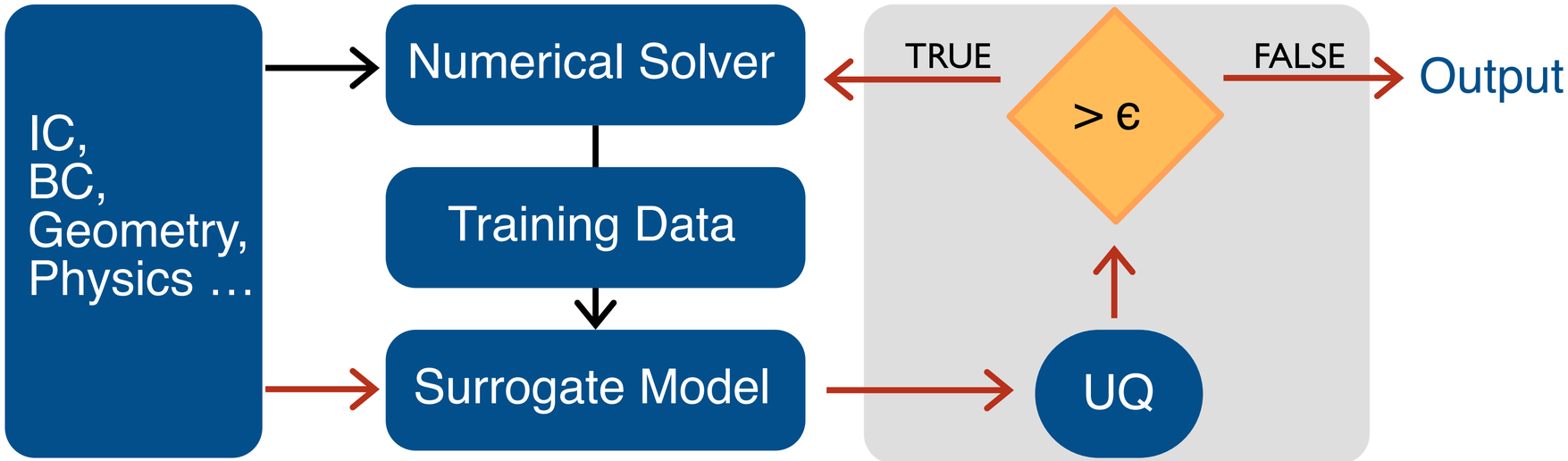
We probably need assisting tools







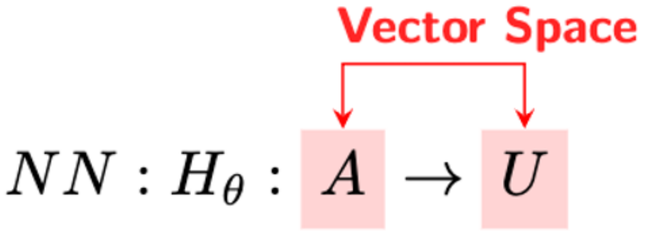
Surrogate Model Ecosystem



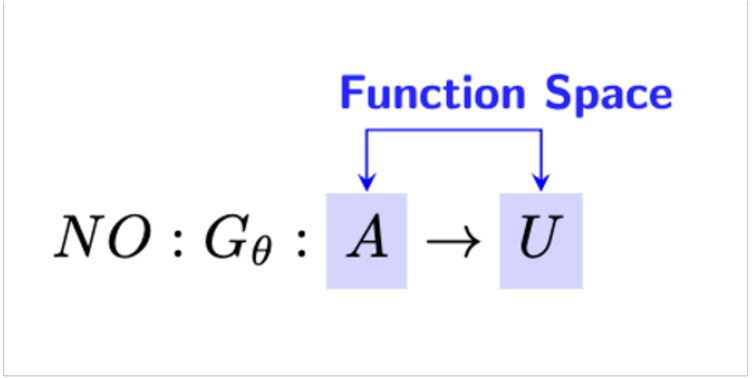
Surrogate Model here refers to a data-driven method such as a neural network

Neural Operators: A primer

- Traditional Neural Networks (MLPs, CNNs ….) map from input vector space to the output vector space, learning the function.

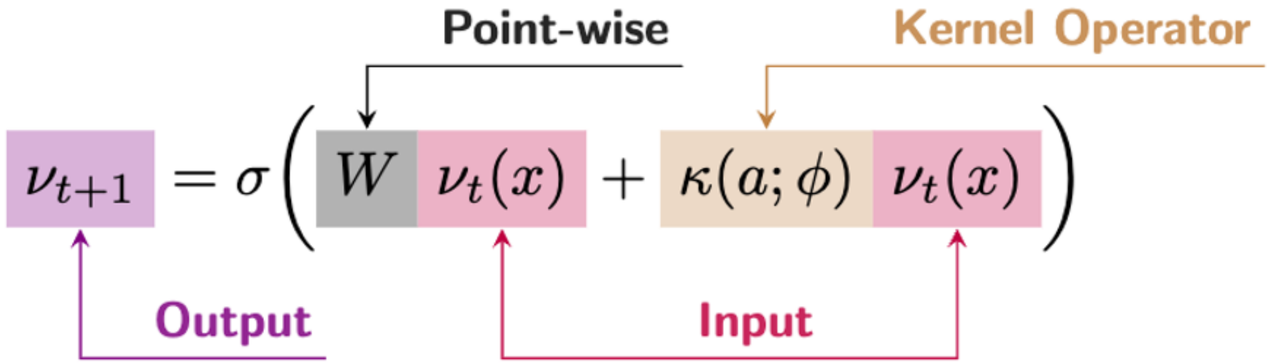


- Neural Operators map the **input function space** to the **output function space**, essentially learning the operator.



Neural Operators: A primer

General Neural Operator Framework:

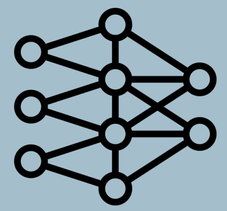


We design different neural operators by our choice of kernel integral transform:

Fourier Transform —> FNO. Laplace Transform —> LNO. Convolutions —> CNO. Self-Attention —> ViT.

Operator Learning

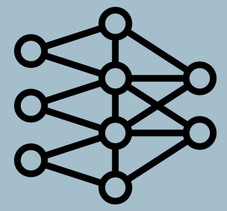
Solution Operator

u^n  u^{n+1}

Autoregressive NO

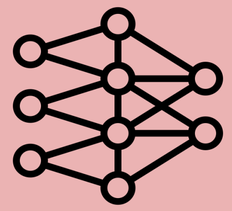
Operator Learning

Solution Operator

u^n  u^{n+1}

Autoregressive NO

Dynamics Operator

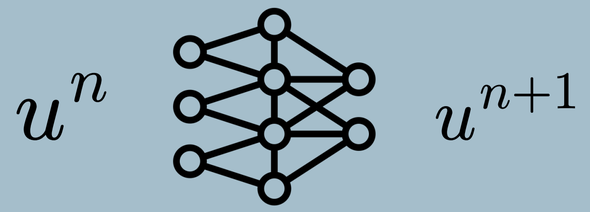
u^n  $\frac{du}{dt}$

Δt

Neural ODE

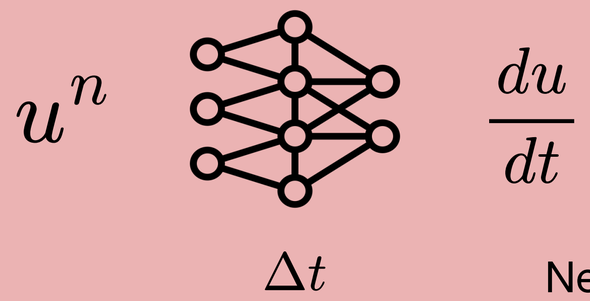
Operator Learning

Solution Operator



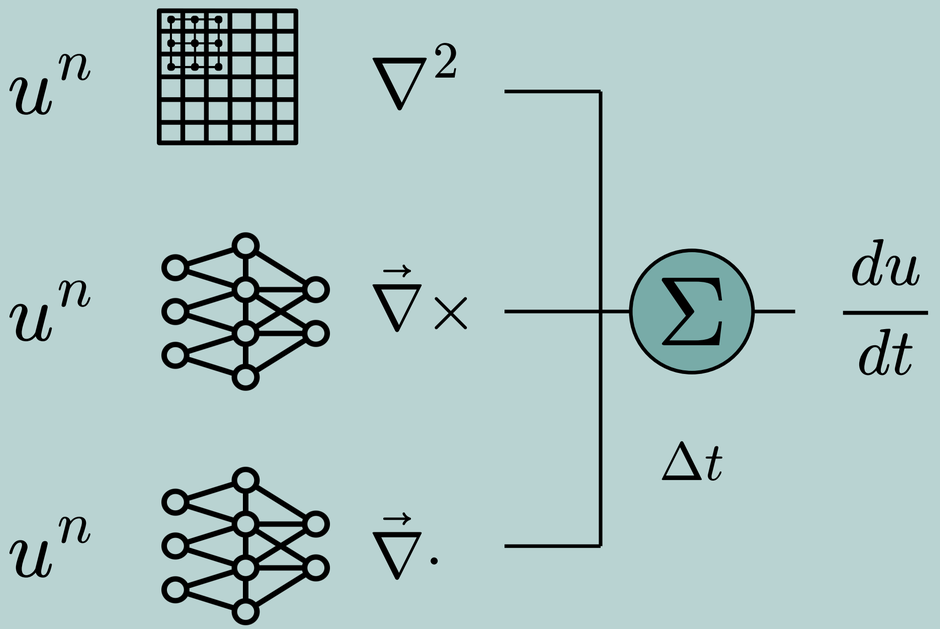
Autoregressive NO

Dynamics Operator



Neural ODE

Physical Operators



OpsSplit

Generalisable Neural Operators:

Mixture-of-Experts structure. Explicitly learn physical operators that can generalise. Modular structure lets operators be added or removed as physics evolves

Continuous in Time:

A neural ODE formulation provides temporal flexibility

Physics-Informed Machine Learning:

Physics consistency is inherently encoded through built-in enforcement of PDE constraints via the operator splitting structure of the prediction regime.

$$u^{n+1} = \left(\sum_{i=1}^{j} \lambda_i \mathbb{FD}_i(u^n) + \sum_{i=1}^{k} \lambda_i \mathbb{NO}_i(u^n) \right) \Delta t + u^n$$

Incompressible Navier-Stokes

Momentum Equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = \nu \nabla^2 \mathbf{v} - \nabla P$$

Incompressible Navier-Stokes

Momentum Equation:

PDE:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = \nu \nabla^2 \mathbf{v} - \nabla P$$

OpsSplit:

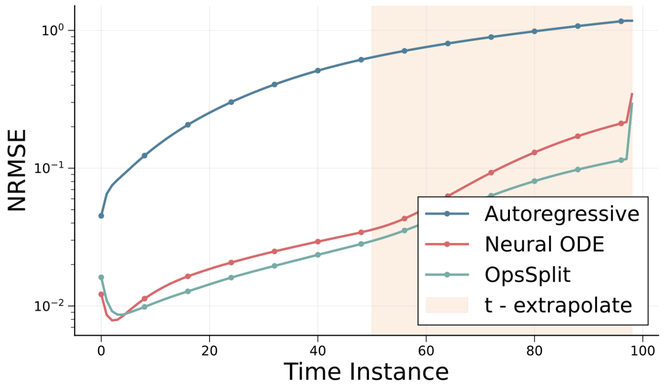
$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\mathbb{NO}_{conv}(\mathbf{v}) + \nu \mathbb{FD}_{\nabla^2}(\mathbf{v})$$

Incompressible Navier-Stokes

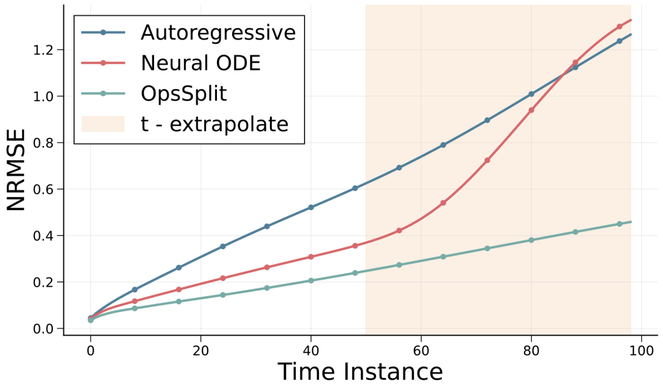
Table

Operator Learning	Model	Time Stepping	OpsSplit	Parameters	Train Time (hrs:mins)	NRMSE			
						Test	t-extrapolate	OOD	OOD+t-extrapolate
Solution	FNO	Autoregressive	False	13437314	1:03	0.3387	0.9126	0.3747	0.9535
Dynamics	FNO	Euler	False	13437314	1:04	**0.0215**	0.1196	0.2209	0.6913
Physical	FNO	Euler	True	13437314	1:09	0.0922	**0.0743**	**0.1465**	**0.3282**
Solution	U-Net	Autoregressive	False	31384322	1:22	0.8346	1.2779	0.8811	1.1851
Dynamics	U-Net	Euler	False	31384322	1:22	0.0425	**0.2039**	0.2634	0.4942
Physical	U-Net	Euler	True	31384322	1:26	**0.0540**	0.2298	**0.2376**	**0.4331**
Solution	ViT	Autoregressive	False	11405510	3:42	0.1282	0.4072	0.7429	0.9607
Dynamics	ViT	Euler	False	11405510	3:43	0.0881	0.3066	0.3916	0.8694
Physical	ViT	Euler	True	11405510	3:45	**0.0810**	**0.2818**	**0.3561**	**0.7752**
Solution	CNO	Autoregressive	False	1420458	9:43	0.4945	0.9278	0.5638	3.2699
Dynamics	CNO	Euler	False	1420458	9:45	0.2986	0.7082	0.4384	0.9049
Physical	CNO	Euler	True	1420458	9:52	**0.2713**	**0.5311**	**0.3541**	**0.7136**
Solution	UNO	Autoregressive	False	1397442	2:23	0.0883	0.2952	0.6003	0.8433
Dynamics	UNO	Euler	False	1397442	2:23	0.0142	**0.1749**	**0.2975**	**0.6210**
Physical	UNO	Euler	True	1397442	2:24	**0.0130**	0.1850	0.3228	0.7127

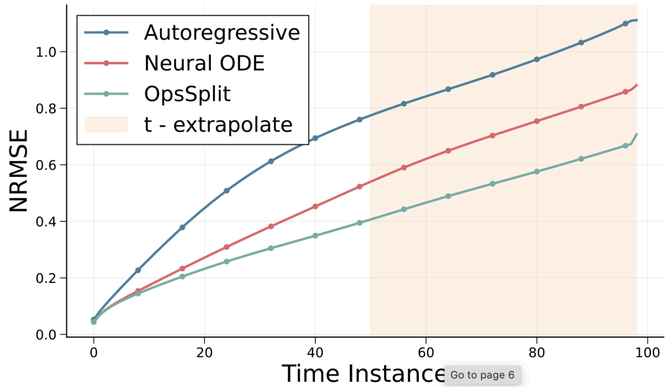
Temporal Error Growth



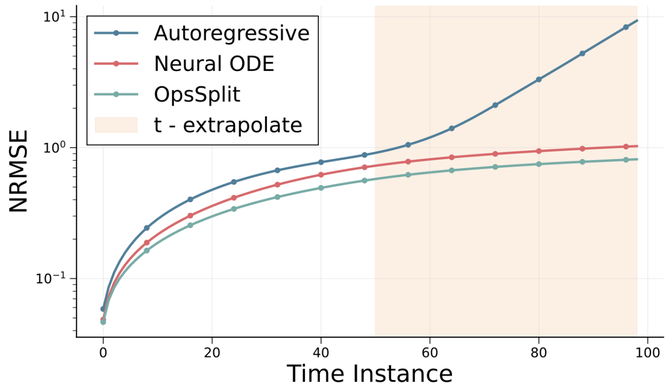
FNO: in-distribution



FNO: out-of-distribution

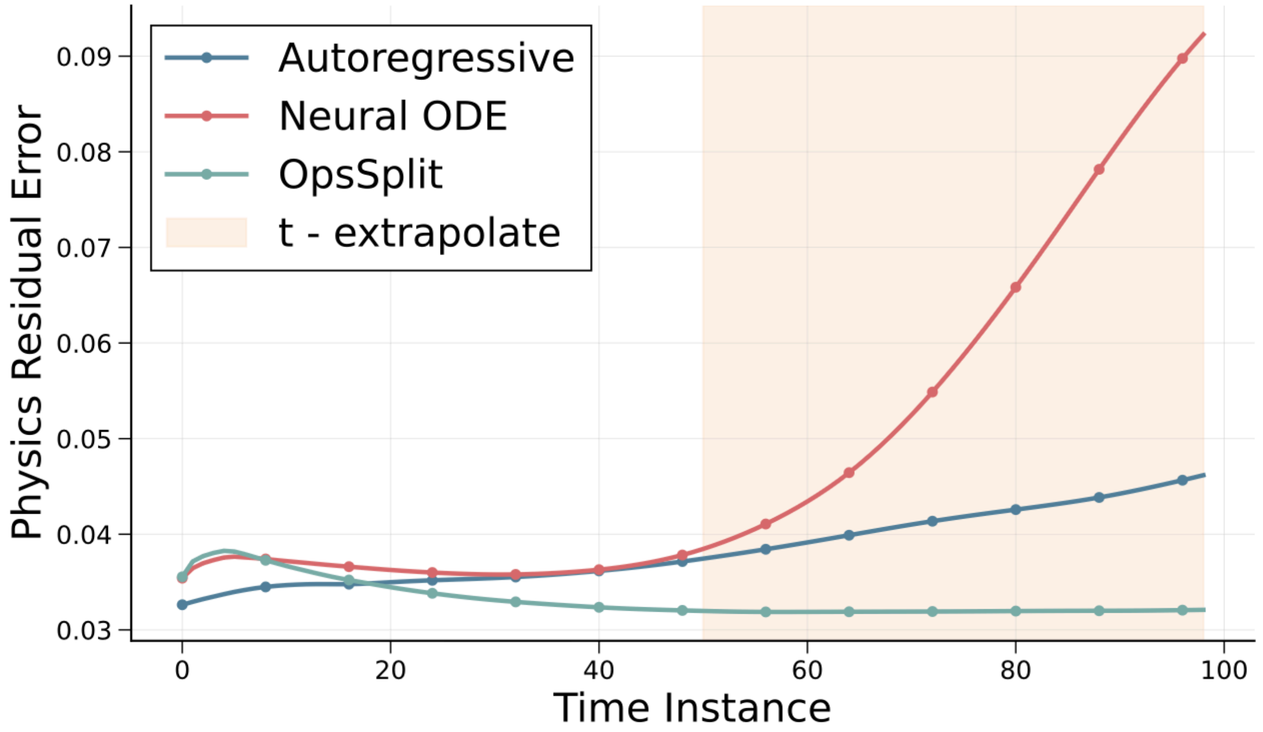


U-Net: in-distribution



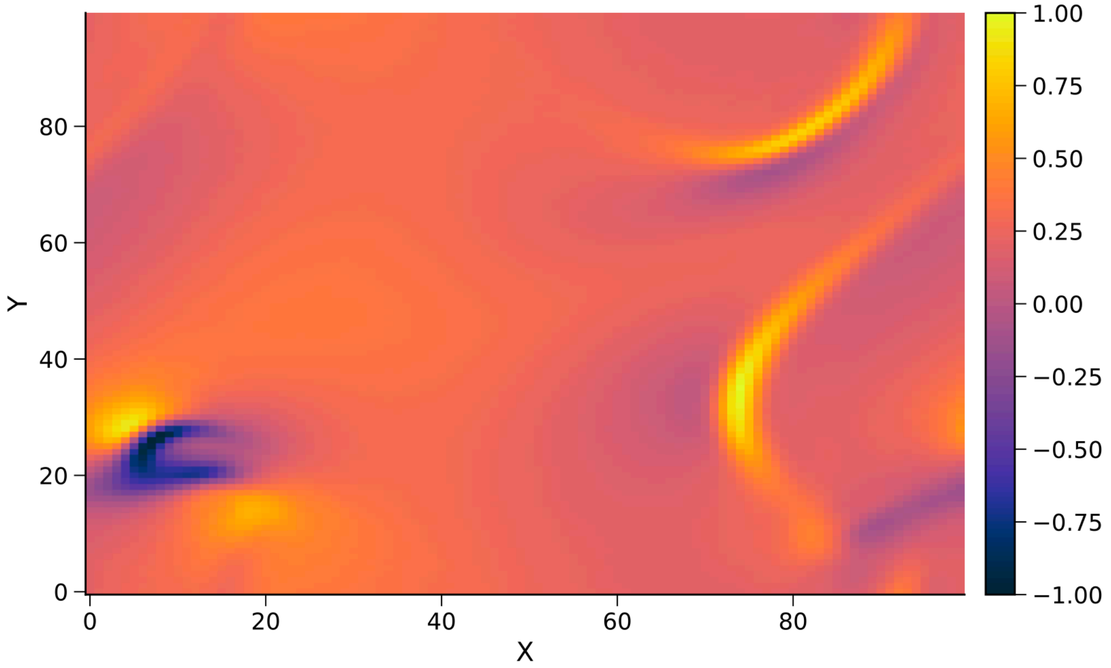
U-Net: out-of-distribution

Physical Consistency

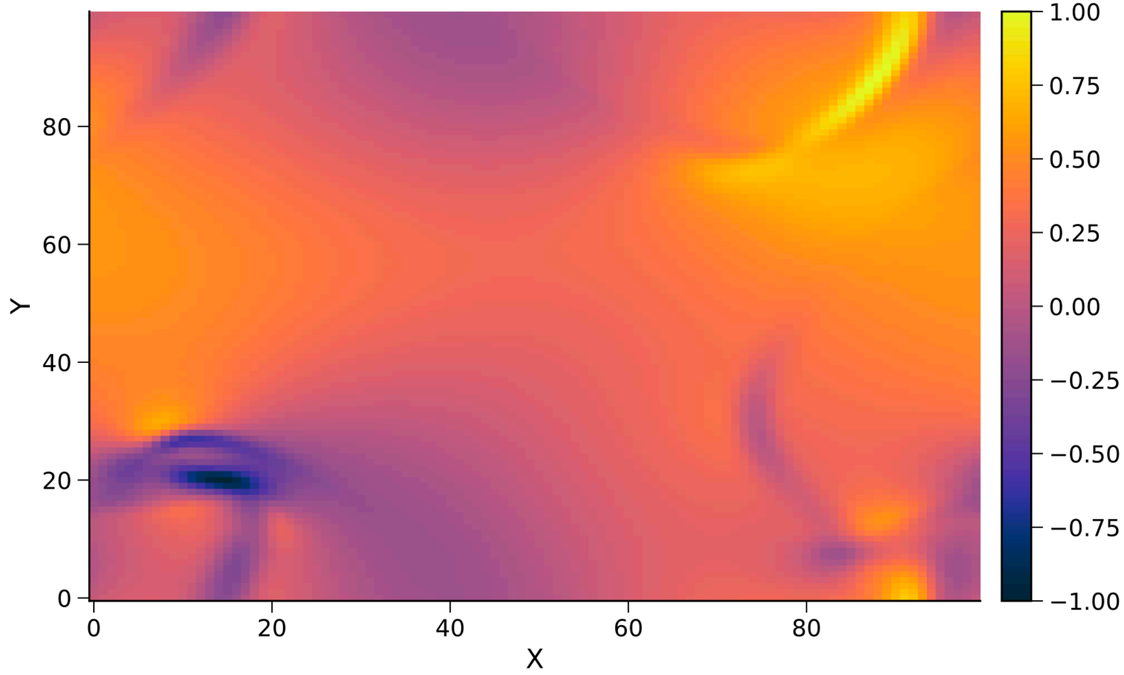


Residual of continuity equation ($\nabla \cdot \mathbf{v}$) across FNO for OOD modelling

Interpretability



Convection Operator: Neural



Convection Operator: Numerical

Compressible Navier-Stokes

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}),$$

(Mass conservation / Continuity)

$$\frac{\partial \mathbf{v}}{\partial t} = -(\mathbf{v} \cdot \nabla)\mathbf{v} - \frac{1}{\rho}\nabla P,$$

(Momentum conservation)

$$\frac{\partial P}{\partial t} = -\mathbf{v} \cdot \nabla P - \gamma P(\nabla \cdot \mathbf{v}).$$

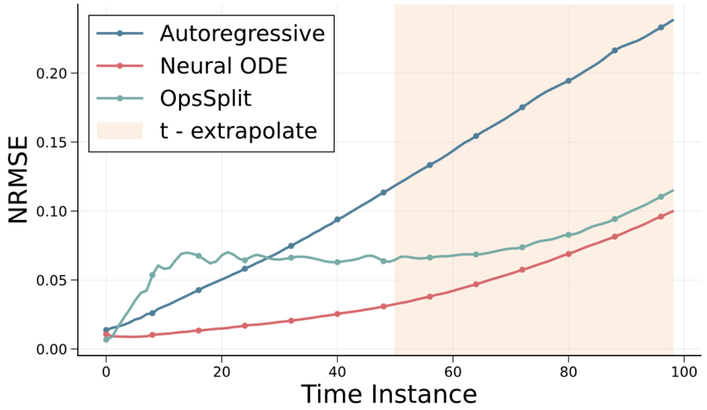
(Pressure evolution)

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\mathbb{NO}_{\nabla}.(\rho, \mathbf{v}),$$

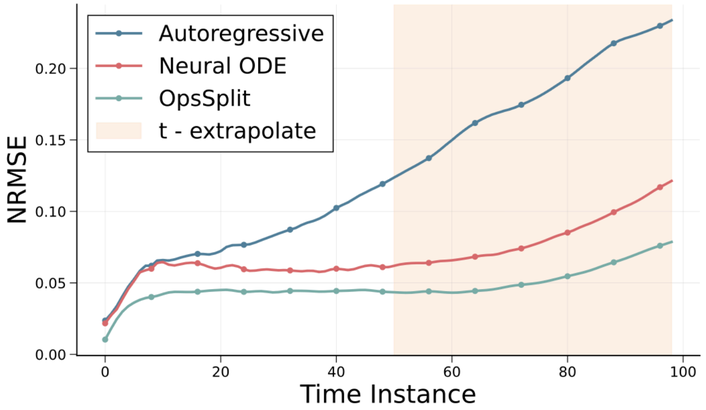
$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\mathbb{NO}_{conv}(\mathbf{v}) - \ln(\rho)\mathbb{FD}_{\nabla}(P),$$

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -\gamma\mathbb{NO}_{\nabla}.(P, \mathbf{v}).$$

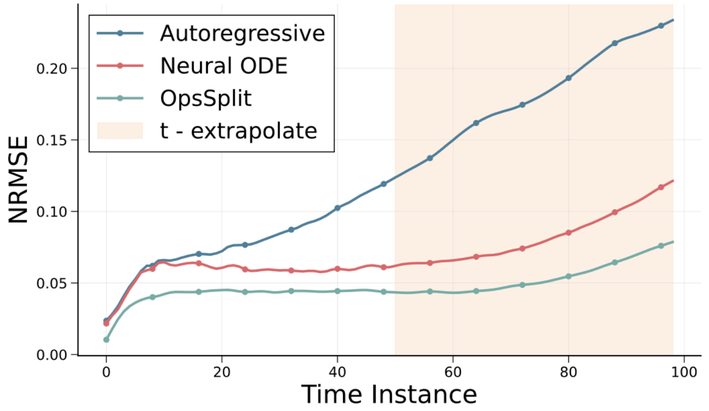
Temporal Error



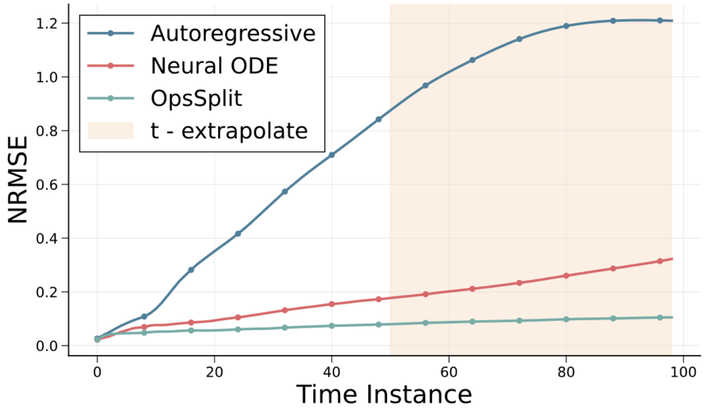
FNO: in-distribution



FNO: out-of-distribution

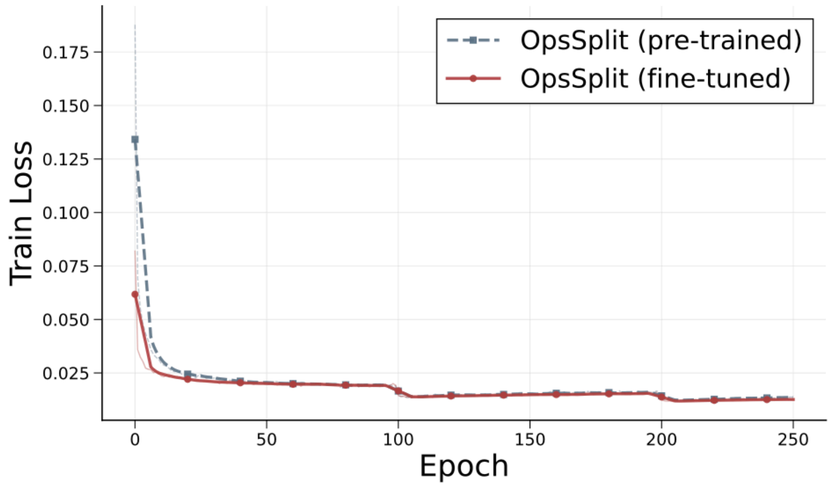


U-Net: in-distribution

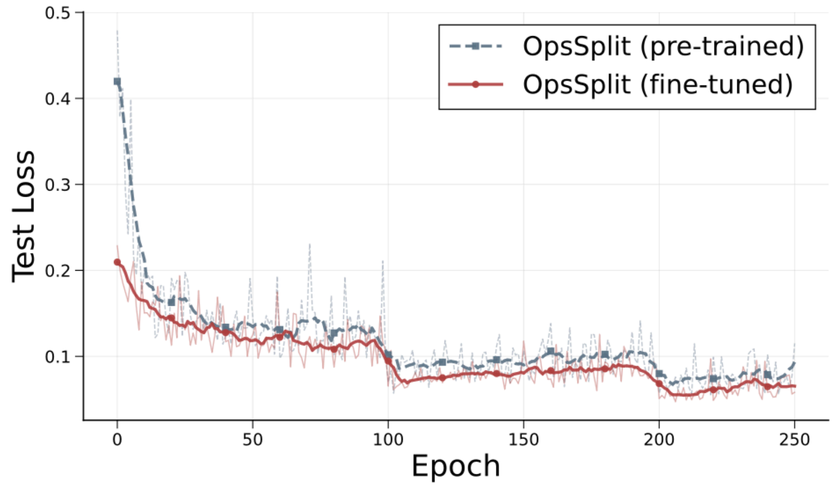


U-Net: out-of-distribution

Fine-tuning



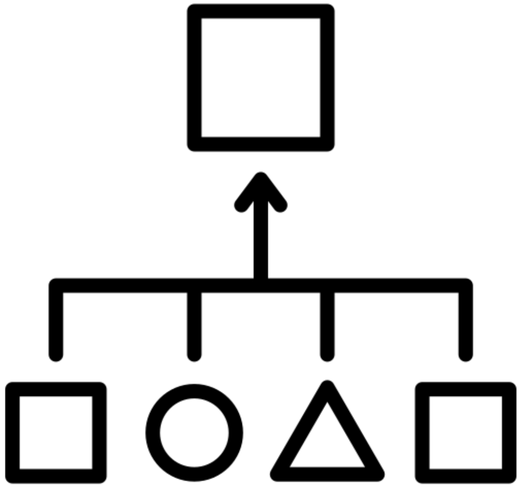
(a) Training loss convergence



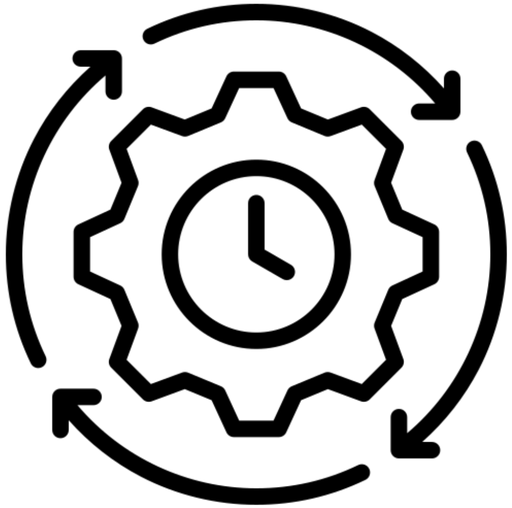
(b) Test loss convergence

Pre-training on compressible Navier–Stokes weights speeds up convergence when fine-tuned for the incompressible Navier–Stokes FNO-based OpsSplit model.

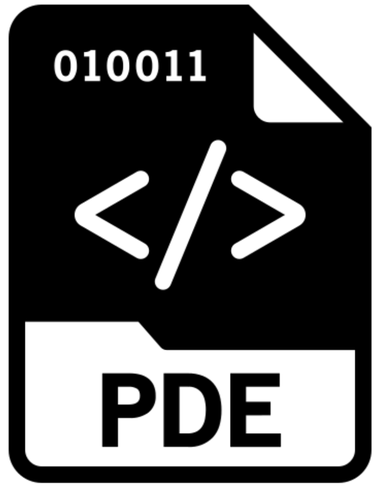
Using Neural Operators to approx. Physical Operators



Generalisable



Temporal



Physics-Informed

Paper :

Gopakumar et al. Learning Physical Operators using Neural Operators